Polynomials — problem 10.41

A. Engel — Problem-Solving Strategies · 1998

Let $f(x) = \left(x^{1958} + x^{1957} + 2\right)^{1959} = a_0 + a_1 x + \cdots + a_n x^n$. Find

$$a_0 - a_1/2 - a_2/2 + a_3 - a_4/2 - a_5/2 + a_6 - \cdots .$$